

Accelerating topological quantum state transfers in optomechanical lattices

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Abstract

This is a simple paragraph at the beginning of the document. A brief introduction about the main subject.

1 Hamiltonian

We consider a 1D small optomechanical lattice containing three cavities and two mechanical resonators. The Hamiltonian of the system is given by

$$\begin{aligned}\hat{H} = & \sum_{n=1,2,3} [\omega_{c,n} \hat{a}_n^\dagger \hat{a}_n + (\Omega_n \hat{a}_n^\dagger e^{-i\omega_d t} + \Omega_n^* \hat{a}_n e^{i\omega_d t})] \\ & + \sum_{l=1,2} [\omega_{m,l} \hat{b}_l^\dagger \hat{b}_l - g_l (\hat{a}_l^\dagger \hat{a}_l - \hat{a}_{l+1}^\dagger \hat{a}_{l+1}) (\hat{b}_l^\dagger + \hat{b}_l) \\ & + G (\hat{a}_{l+1}^\dagger \hat{a}_l + \hat{a}_l^\dagger \hat{a}_{l+1})].\end{aligned}\quad (1)$$

Where $\hat{a}_n$ and $\hat{b}_l$ are the annihilation operators of the cavity and mechanical resonator, respectively. $\omega_{c,n}$ and $\omega_{m,l}$ are the frequencies of the cavity and mechanical resonator, respectively. Ω_n is the driving strength of the cavity. ω_d is the frequency of the driving field. g_l and G are the coupling strengths between the cavity and mechanical resonator, and between the cavities, respectively.

We using driving frequency ω_d , to perform the rotating transformation, the hamiltonian can be written as (n=1,l=1):

$$\begin{aligned}\hat{H}_{n=1,l=1} = & (\omega_{c,1} - \omega_d) \hat{a}_1^\dagger \hat{a}_1 + \Omega_1 \hat{a}_1^\dagger + \Omega_1^* \hat{a}_1 \\ & + \omega_{m,1} \hat{b}_1^\dagger \hat{b}_1 - g_1 (\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2) (\hat{b}_1^\dagger + \hat{b}_1) \\ & + G (\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2).\end{aligned}\quad (2)$$

Similarly, the full Hamiltonian can be written as:

$$\hat{H}_R = \sum_{n=1,2,3} [\Delta_{c,n} \hat{a}_n^\dagger \hat{a}_n + (\Omega_n \hat{a}_n^\dagger + \Omega_n^* \hat{a}_n)]$$

$$\begin{aligned}
& + \sum_{l=1,2} [\omega_{m,l} \hat{b}_l^\dagger \hat{b}_l - g_l (\hat{a}_l^\dagger \hat{a}_l - \hat{a}_{l+1}^\dagger \hat{a}_{l+1}) (\hat{b}_l^\dagger + \hat{b}_l) \\
& + G(\hat{a}_{l+1}^\dagger \hat{a}_n + \hat{a}_l^\dagger \hat{a}_{l+1})].
\end{aligned} \tag{3}$$

2 Standard linearization process

Optomechanical system could be described by quantum Langevin equations:

$$\dot{\mathcal{O}} = \frac{i}{\hbar} [H, \mathcal{O}] - \frac{\gamma}{2} \mathcal{O}. \tag{4}$$

We can obtain the following quantum Langevin equations for the operators $\hat{a}_n$ and $\hat{b}_l$:

$$\begin{aligned}
\dot{\hat{a}}_1 &= -[\gamma_1 + i\Delta_{c,1} - ig_1(\hat{b}_1^\dagger + \hat{b}_1)]\hat{a}_1 - i(\hat{a}_2 - \Omega_1) \\
\dot{\hat{a}}_2 &= -[\gamma_2 + i\Delta_{c,2} + ig_1(\hat{b}_1^\dagger + \hat{b}_1) - ig_2(\hat{b}_2^\dagger + \hat{b}_2)]\hat{a}_2 - i(\hat{a}_1 - \Omega_2 + \hat{a}_3) \\
\dot{\hat{a}}_3 &= -[\gamma_3 + i\Delta_{c,3} + ig_2(\hat{b}_2^\dagger + \hat{b}_2)]\hat{a}_3 - i(\hat{a}_2 - \Omega_3) \\
\dot{\hat{b}}_1 &= -[\gamma_4 + i\omega_{m,1}\hat{b}_1 - ig_1(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2)] \\
\dot{\hat{b}}_2 &= -[\gamma_5 + i\omega_{m,2}\hat{b}_2 - ig_2(\hat{a}_2^\dagger \hat{a}_2 - \hat{a}_3^\dagger \hat{a}_3)].
\end{aligned} \tag{5}$$

Rewriting the operators as $\hat{a}_n = \langle \hat{a}_n \rangle + \delta \hat{a}_n = \alpha_n + \delta \hat{a}_n$ and $\hat{b}_n = \langle \hat{b}_n \rangle + \delta \hat{b}_n = \beta_n + \delta \hat{b}_n$:

$$\begin{aligned}
\dot{\alpha}_1 + \delta \dot{\hat{a}}_1 &= -[\gamma_1 + i\Delta_{c,1} - ig_1(\beta_1^* + \delta \hat{b}_1^\dagger + \beta_1 + \delta \hat{b}_1)](\alpha_1 + \delta \hat{a}_1) \\
&\quad - i(\alpha_2 + \delta \hat{a}_2 - \Omega_1) \\
\dot{\alpha}_2 + \delta \dot{\hat{a}}_2 &= -[\gamma_2 + i\Delta_{c,2} + ig_1(\beta_1^* + \delta \hat{b}_1^\dagger + \beta_1 + \delta \hat{b}_1) \\
&\quad - ig_2(\beta_2^* + \delta \hat{b}_2^\dagger + \beta_2 + \delta \hat{b}_2)](\alpha_2 + \delta \hat{a}_2) \\
&\quad - i(\alpha_2 + \delta \hat{a}_1 - \Omega_2 + \alpha_3 + \delta \hat{a}_3) \\
\dot{\alpha}_3 + \delta \dot{\hat{a}}_3 &= -[\gamma_3 + i\Delta_{c,3} + ig_2(\beta_2^* + \delta \hat{b}_2^\dagger + \beta_2 + \delta \hat{b}_2)](\alpha_3 + \delta \hat{a}_3) \\
&\quad - i(\alpha_3 + \delta \hat{a}_3 - \Omega_3) \\
\dot{\beta}_1 + \delta \dot{\hat{b}}_1 &= -[\gamma_4 + i\omega_{m,1}(\beta_1 + \delta \hat{b}_1) - ig_1(\alpha_1^* + \delta \hat{a}_1^\dagger)(\alpha_1 + \delta \hat{a}_1) \\
&\quad - ig_1(\alpha_2^* + \delta \hat{a}_2^\dagger)(\alpha_2 + \delta \hat{a}_2)] \\
\dot{\beta}_2 + \delta \dot{\hat{b}}_2 &= -[\gamma_5 + i\omega_{m,2}(\beta_2 + \delta \hat{b}_2) - ig_2(\alpha_2^* + \delta \hat{a}_2^\dagger)(\alpha_2 + \delta \hat{a}_2) \\
&\quad - ig_2(\alpha_3^* + \delta \hat{a}_3^\dagger)(\alpha_3 + \delta \hat{a}_3)].
\end{aligned} \tag{6}$$

By defining the system operator as its mean plus the fluctuation of the operator, the second-order terms are ignored. We can obtain the Langevin equations corresponding to the operator mean:

$$\begin{aligned}
\dot{\alpha}_1 &= -[\gamma_1 + i\Delta_{c,1} - ig_1(\beta_1^* + \beta_1)]\alpha_1 - i(\alpha_2 - \Omega_1) \\
\dot{\alpha}_2 &= -[\gamma_2 + i\Delta_{c,2} + ig_1(\beta_1^* + \beta_1) - ig_2(\beta_2^* + \beta_2)]\alpha_2 - i(\alpha_1 - \Omega_2 + \alpha_3)
\end{aligned}$$

$$\begin{aligned}
\dot{\alpha}_3 &= -[\gamma_3 + i\Delta_{c,3} + ig_2(\beta_2^* + \beta_2)]\alpha_3 - i(\alpha_2 - \Omega_3) \\
\dot{\beta}_1 &= -[\gamma_4 + i\omega_{m,1}\beta_1 - ig_1(\alpha_1^*\alpha_1 - \alpha_2^*\alpha_2)] \\
\dot{\beta}_2 &= -[\gamma_5 + i\omega_{m,2}\beta_2 - ig_2(\alpha_2^*\alpha_2 - \alpha_3^*\alpha_3)].
\end{aligned} \tag{7}$$

and the Langevin equations corresponding to the fluctuation operator:

$$\begin{aligned}
\delta\dot{\hat{a}}_1 &= -[\gamma_1 + i\Delta_{c,1} - ig_1(\beta_1^* + \beta_1)]\delta\hat{a}_1 - ig_1(\delta\hat{b}_1^\dagger + \delta\hat{b}_1)\alpha_1 - i\delta\hat{a}_2 \\
\delta\dot{\hat{a}}_2 &= -[\gamma_2 + i\Delta_{c,2} + ig_1(\beta_1^* + \beta_1) - ig_2(\beta_2^* + \beta_2)]\delta\hat{a}_2 \\
&\quad + [ig_1(\delta\hat{b}_1^\dagger + \delta\hat{b}_1) - ig_2(\delta\hat{b}_2^\dagger + \delta\hat{b}_2)]\alpha_2 - i\delta\hat{a}_1 - i\delta\hat{a}_3 \\
\delta\dot{\hat{a}}_3 &= -[\gamma_3 + i\Delta_{c,3} + ig_2(\beta_2^* + \beta_2)]\delta\hat{a}_3 - ig_2(\delta\hat{b}_2^\dagger + \delta\hat{b}_2)\alpha_3 - i\delta\hat{a}_2 \\
\delta\dot{\hat{b}}_1 &= -[\gamma_4 + i\omega_{m,1}\delta\hat{b}_1 \\
&\quad - ig_1(\alpha_1^*\alpha_1 - \alpha_2^*\alpha_2 + \alpha_1^\dagger\delta\hat{a}_1 + \alpha_1\delta\hat{a}_1^\dagger - \alpha_2^\dagger\delta\hat{a}_2 - \alpha_2\delta\hat{a}_2^\dagger)] \\
\delta\dot{\hat{b}}_2 &= -[\gamma_5 + i\omega_{m,2}\delta\hat{b}_2 \\
&\quad - ig_2(\alpha_2^*\alpha_2 - \alpha_3^*\alpha_3 + \alpha_2^\dagger\delta\hat{a}_2 + \alpha_2\delta\hat{a}_2^\dagger - \alpha_3^\dagger\delta\hat{a}_3 - \alpha_3\delta\hat{a}_3^\dagger)].
\end{aligned} \tag{8}$$

After linearization, the system Hamiltonian corresponding to the Langevin equations of the fluctuation operator is obtained as

$$\begin{aligned}
\hat{H}_L &= \sum_{n=1,2,3} [\Delta_{c,n}\hat{a}_n^\dagger\hat{a}_n] + \sum_{l=1,2} [\omega_{m,l}\hat{b}_l^\dagger\hat{b}_l - g_l(\alpha_l^*\hat{a}_l + \alpha_l\hat{a}_l^\dagger - \alpha_{l+1}^*\hat{a}_{l+1} \\
&\quad - \alpha_{l+1}\hat{a}_{l+1}^\dagger)(\hat{b}_l^\dagger + \hat{b}_l) + G(a_{l+1}^\dagger\hat{a}_l + a_l^\dagger\hat{a}_{l+1})].
\end{aligned} \tag{9}$$

where $\Delta_{c,n}$ are the effective detuning of the cavity and the driving fields originating from optomechanical coupling, respectively. Then we perform rotating transformation with respect to $\Delta_{c,n}\hat{a}_n^\dagger\hat{a}_n$ and $\omega_{m,l}\hat{b}_l^\dagger\hat{b}_l$ to obtain the Hamiltonian in the rotating frame:

$$\begin{aligned}
\hat{H}_L^{(1)} &= \sum_{n=1,2} [-G'_n\hat{a}_n^\dagger\hat{b}_n e^{i(\Delta_{c,n}-\omega_{m,n})t} \\
&\quad - G'_n\hat{a}_n^\dagger\hat{b}_n^\dagger e^{i(\Delta_{c,n}+\omega_{m,n})t} \\
&\quad + G'_{n+1}\hat{a}_{n+1}^\dagger\hat{b}_n e^{i(\Delta_{c,n+1}-\omega_{m,n})t} \\
&\quad + G'_{n+1}\hat{a}_{n+1}^\dagger\hat{b}_n^\dagger e^{i(\Delta_{c,n+1}+\omega_{m,n})t} + Ga_{n+1}^\dagger\hat{a}_n] + \text{H.c.}
\end{aligned} \tag{10}$$

where $G'_n = g_n\alpha_n$ are the effective optomechanical coupling of the cavity. Under red-detuning resonant condition, the counter rotating wave terms can be removed. The Hamiltonian becomes:

$$\hat{H} = \sum_{n=1,2} [-G'_n\hat{a}_n^\dagger\hat{b}_n + G'_{n+1}\hat{a}_{n+1}^\dagger\hat{b}_n + Ga_{n+1}^\dagger\hat{a}_n] + \text{H.c.} \tag{11}$$

When $G = 0$, the above Hamiltonian is equivalent to a standard SSH-type tight-binding Hamiltonian possessing only the nearest-neighboring couplings,